

Exploring Norwegian company register data

CS4010: Data Management Course

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Abstract

Keywords:

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1 Introduction

1.1 Problem Statement

Financial performance is of central interest for entrepreneurs, investors, and managers alike. Financial sustainability also directly relates to the United Nations Sustainable Development Goals (SDGs), most importantly SDG 8 “Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all” [1]. SDG 8 and financial sustainability are also necessary to reach the targets set out in other SDGs [2].

Insights into company financial metrics and relationships between financial performance and other factors might as such be useful for company stakeholders. Two examples of relevant factors are industry sector, represented by NACE code [3], and geographical indicators such as company location. Combinations of factors can also be relevant

1.2 Project objective

The aim of this project is to provide managers, investors, and other company stakeholders with insights that enable them to make better business decisions and reach their goals for their business or investment by exploring official Norwegian register data for all 1,171,373 Norwegian companies registered per 25th August 2026 [4]. For this business value to be more accessible, insights will be presented visually where appropriate, using Microsoft Power BI as the main platform for such visualisations.

1.3 Assumptions and limitations

We focus on limited companies (“Aksjeselskap” - “AS”) [5] [6] and include solely Norwegian companies. These two limitations present a constraint for stakeholders seeking to gain insights beyond the Norwegian population of limited companies.

Another limitation stems from the API used to retrieve financial reports. The open API used only provides one year of financial reports per company, and time series of financial data is thus not part of the data set used in this project [7].

Another limitation also relates to the financial data. For companies established in the most recent years, financial data will typically not be available, due to the annual nature of the financial reports and the delay that compiling the financial accounts imposes. By law, the financial reports are to be submitted no later than the 31th of July the following year [8]. Late or missing submittals also present a limitation in the data set. Per 31st August 2026 The Brønnøysund Register Centre had 2,837 active announcements for companies risking forced dissolution due to missing submission of annual accounts [9].

1.4 Acknowledgements

The project group consisted of three members. 8634 focused on retrieving and processing the company and financial data. 8583 mainly worked with data analysis and the Power BI reports. 8615 retrieved and processed the population data. All members contributed to writing the project report itself. Artificial intelligence (AI) was used as a discussion partner in this project. Primarily, it was used to refine and troubleshoot Docker and Python commands, which improved the pace and extended the scope of the data analysis. AI also supported proofreading by improving grammar, sentence structure, and linguistic clarity. All arguments, findings, and theoretical reflections are the author's own work, as well as any errors.

2 Theoretical Background

The data sets used in this project exhibit the volume, variety and, to a lesser extent, the velocity characteristics commonly used to describe big data [10]: More than 1.1 million records across multiple formats and sources, ingested through re-runnable incremental scripts rather than one-off data loads. These characteristics influenced which tools were appropriate for this specific project.

The company register contains nested and optional information, such as multiple addresses, roles, and sub-entities per company. Therefore, a document-oriented database was more preferable to a traditional relational one. MongoDB was selected because it can store records in a flexible document structure without requiring that every field be defined in advance. This was appropriate for data sets where some fields are missing or occur only in certain records.

Apache Spark was chosen because the project combines several data sources and requires several large joins, transformations, and aggregations. Spark supports different input formats through a common processing framework, making it suitable for combining company and financial data, and for analysing results by industry, municipality, and company status. Parquet was chosen as the main analytical storage format because it is a column-oriented format that works well with Spark.

Power BI was chosen to present the results, as the project is intended for managers, investors, and stakeholders who may not work directly with databases or programming tools. It provides an accessible way to explore patterns in profitability, industries, and geographical differences.

3 Literature Review

Try to do some literature review and find any similar system is existing. If yes, write how your project is different than them. If not, list similar work.

Several studies have looked at how open government data (OGD) can generate business value.

Wahlstrøm (2022)

4 Research Methodology

To gain the aimed insights in this project, multiple big data technologies were used. Designing and implementing a robust data pipeline to retrieve and process data was emphasised. This enabled us to expand the project value beyond insights from the specific data sets to also provide a methodology and a near production-ready big data environment.

The first step was data identification and retrieval. Data about organisations was downloaded in JSON format [11], containing register information on all Norwegian Companies [4]. At the date of download, this equalled 1,171,373 companies. The file size of the compressed downloadable JSON file was around 200 MB (uncompressed 2 GB), and download was completed in a few seconds.

The same data set is also available in CSV and Excel format, as well as through an

API. While CSV might seem like the most appropriate format at first glance, due to the smaller file size and fast processing speed, JSON was still chosen. The main reasons are the nested structure of the data, CSV cells containing line breaks and quotation marks resulting in additional data cleaning, and finally, type fidelity. The subsequent inclusion of additional data sources as well as further processing down the data pipeline suggests that storing the data in MongoDB might be more appropriate than working with the JSON file directly, allowing data to be stored in the natural shape, and supporting fast initial queries.

Retrieving the same data via the The Brønnøysund Register Centre’s API might have been chosen in a production scenario. However, as we are retrieving another data set via an API, the file download method was chosen for the data set on organisations for the sake of demonstrating different approaches to data retrieval.

Figure 4.1 depicts a high-level overview of the data processing pipeline and data flow.

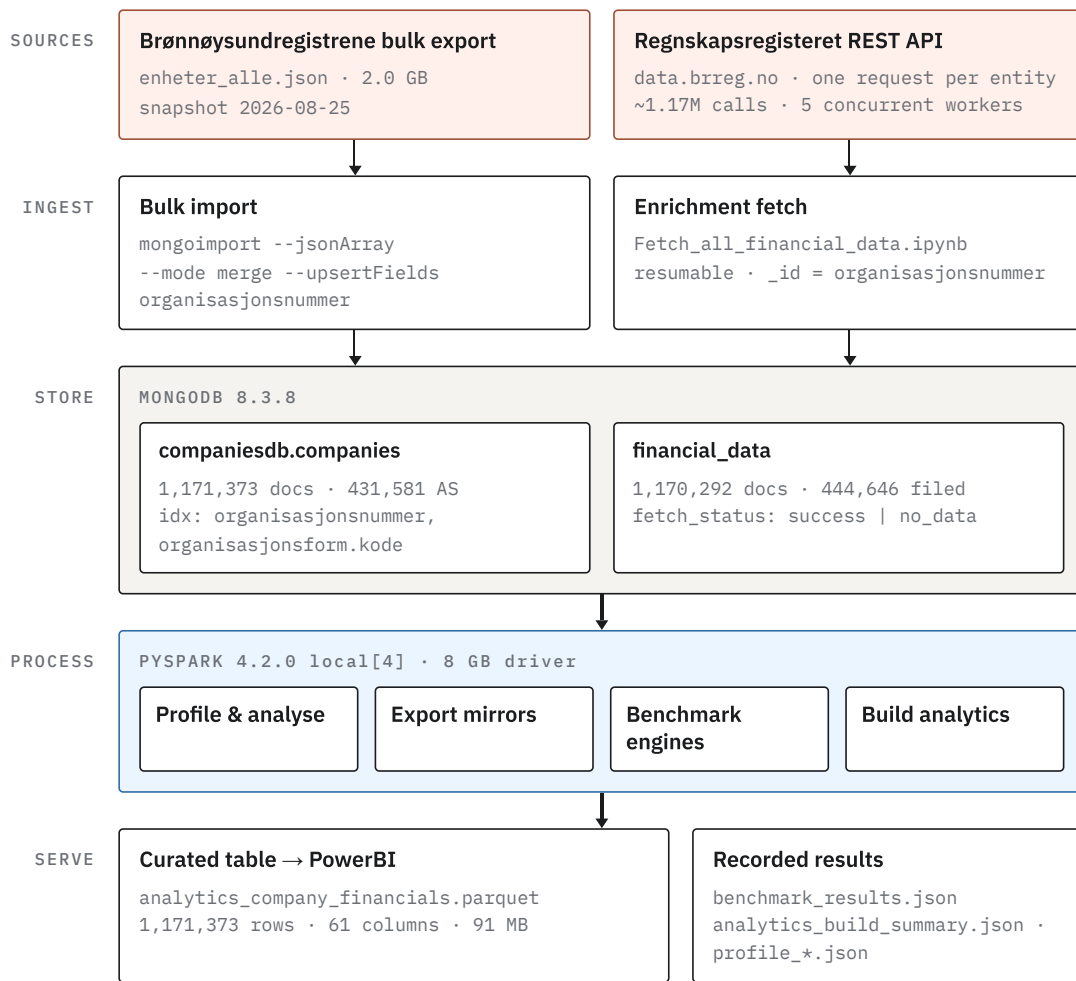


Figure 4.1: Illustration of data flow and processing pipeline

Note. Source: Own illustration assisted by Claude Opus.

The other data set in question is “Nøkkeltall fra Årsregnskapet”, made available by The Norwegian Digitalisation Agency [7]. This data set contains the most recently submitted annual financial report for each company, if such accounts are available. As this API requires an organisation number as input and returns the financial report for one single company, we created a Python script that looped through all companies in the company data set, and tried to fetch financial data for all of these. The script was created and run in a Jupyter notebook. We limited ourselves to five concurrent requests to not risk exhausting the API or being blocked, and the script plateaued at fetching around 30 financial records per second, retrieving all available records in the course of an overnight run. An interesting finding is the HTTP 500 responses from the API. While only occurring

for a small subset of the requests, the ratio is persistent across days and spread across all 16 legal forms. Table 4.1 shows descriptive statistics for the initial runs of this script.

Table 4.1: Retrieval of financial reports

Outcome	Count	% of register
Organisation numbers requested	1,171,373	100.00
Responses recorded	1,170,292	99.91
HTTP 200 – statement returned	444,646	37.96
HTTP 404 – no statement filed	725,646	61.95
HTTP 500 – unresolved after retries	1,081	0.09

Note. Own calculations based on script logs and output.

While the company data was imported to a MongoDB database collection with a docker command, the financial data was inserted directly into a second MongoDB collection as part of the Python script. The company data could have been transformed directly to Parquet files, although MongoDB enhances schema creation and is necessary for several of the benchmarks also for the company data.

The financial data, however, is another story. First, when running five workers in parallel, collecting 1.17M responses to one or more Parquet files would require using a buffer, which could be lost if interrupted. Second, inserting requests to MongoDB rather than Parquet files simplifies robustness and resumability of the retrieval script. Third, initial storage in MongoDB enhances schema creation.

While the Parquet files contain the schema for the data in the footer, and a separate schema is thus not needed when *reading* the Parquet files, we need a schema to *write* those Parquet files. We automated the creation of this schema by performing a full profiling pass of all records stored in MongoDB. While reading though all the rows in the source might seem unnecessary, several fields only have value for a subset of companies - the most extreme being “tvangsopplostPgaManglendeDagligLederDato” having data in only one of 1,171,373 records. A full profiling pass ensures a complete and correct schema. Our script also profiles both the source JSON file and the MongoDB company collection, which might be considered redundant, but this acts as an extra verification that the ingestion of data from JSON to MongoDB is correct.

The script that imports company data can be re-run at any time, and new companies in the file will be added to the MongoDB collection. Using the “merge” option for the

“mongoimport” command also updates fields that have changed for existing companies. Updates occur at the top level for each field.

Using the merge option also enables us to preserve companies that no longer exist in the source data set, while including new companies on import re-runs. We also output the “mongoimport” log to a separate log file to preserve an audit trail. The import could be expanded by adding timestamps for all changes, tracking deleted companies as well as storing historical data for each field in MongoDB, depending on the system’s requirements. Similarly, the script to fetch financial data will try to fetch financial data only for those companies where financial data is currently missing, and add any new financial reports to the MongoDB collection. Companies whose requests responded with an error code will also be attempted to be re-fetched. The script is configurable as to what kinds of companies should be retried to fetch information on. The default setting is to retry for limited companies only, as sole proprietors and associations are mostly not required to submit financial reports [8], and trying to re-fetch these would result in unnecessary overhead. The subsequent steps of the pipeline follow the pattern of appending new information, including benchmarks, as well as the Parquet export used as the basis for Power BI analysis.

We used Python scripts in a Jupyter notebook to analyse the data and display descriptive statistics. Table 4.2 depicts select descriptive statistics for the combined data set.

The next step in the pipeline is exporting data to Parquet. Parquet is used as the main data source for Spark as it is considered to be highly efficient and a preferred data source for Spark[12]. An additional export to NDJSON for benchmark purposes is also performed.

In a live environment, some further enhancements would typically be implemented, namely automation of the different tasks in the steps, including a nightly fetch of new/updated data and subsequent data processing to always provide the most comprehensive and up-dated data set to the end users in Power BI visualisations and other channels.

Some steps in our processing pipeline would on the other hand typically be excluded from a final production installation. Primarily this is related to the benchmarks presented in this project. We suggest the benchmarks themselves would not be included in a live

Table 4.2: Company and financial statistics

Item	Count	%
<i>Register (% of all entities)</i>		
Registered entities	1,171,373	100.00
Entities with a financial statement	444,646	37.96
Founding date missing	496,063	42.35
<i>Statement coverage by legal form (% of form)</i>		
Aksjeselskap (AS)	403,782 / 431,581	93.56
Borettslag (BRL)	9,995 / 10,244	97.57
Eierseksjonssameie (ESEK)	10,324 / 33,212	31.09
Norske registrert utenlandsk foretak (NUF)	3,545 / 25,838	13.72
Forening/lag/innretning (FLI)	2,810 / 128,677	2.18
Enkeltpersonforetak (ENK)	3,266 / 461,154	0.71
<i>The 444,646 statements (% of filed)</i>		
Fiscal year 2025	417,897	93.98
Reported in NOK	443,462	99.73
Full 12-month period	425,549	95.71
Classified as small enterprise	434,468	97.71
Parent company	57,002	12.82
Operating margin computable	293,156	65.93
No operating revenue reported	150,232	33.79

Note. Own calculations based on data from The Brønnøysund Register Centre [4] and The Norwegian Digitalisation Agency [7].

setup, or that they are reduced to only include benchmarks of the components actually used and necessary for the data analysis. Several steps in the pipeline are also redundant given absence of or reduced benchmarks. The two main redundant steps are Apache Spark reading directly from a ~ 2 GB JSON file, and the export from MongoDB to NDJSON, both of which are only related to the benchmarks.

Table 4.3 displays main workload results from the benchmarks, and supports the selection of Spark + Parquet as the chosen path for data processing.

An effect of query formulation is shown in variants A1 and A2, which execute on the same engine against the same collections in the same session and differ only in how the aggregation is expressed: A1 drives from the financial collection and restricts to limited companies after joining, while A2 restricts first and then probes the financial collection on its primary key.

Engine choice is another effect, measured as the ratio between the fastest MongoDB and the fastest Spark result on each workload so that neither side is penalised for a

Table 4.3: Query engine benchmark: median execution time in seconds by workload

Variant			Median execution time (s)		
			W1	W3	W4
A1	MongoDB, financial-first	MongoDB	42.00	— ^a	20.39
A2	MongoDB, AS-first	MongoDB	7.38	— ^a	17.37
A	MongoDB	MongoDB	— ^a	0.43	— ^a
B	Spark	MongoDB connector	10.75	1.50	19.30
C	Spark	Parquet	2.71	1.39	3.69
D	Spark	JSON / NDJSON	35.99	31.16	44.78
Rows returned			2	11	45,033

W1: selective join, restricted to AS, counted by fetch status. W3: unindexed predicate, no join. W4: wide read plus aggregation over industry code and municipality.

^a W3 contains no join, so the two MongoDB formulations are identical there and a single variant is reported. W1 and W4 carry both.

Note. Own measurements.

poorly expressed query. Spark reading Parquet is 2.73 times faster than MongoDB on W1 and 4.71 times faster on W4, but MongoDB is 3.22 times faster on W3. W3 performs an unindexed scan returning eleven rows, and at this data volume Spark’s fixed job-scheduling cost is not recovered by the parallel scan.

Running the same curated-table build at thread counts of 1, 2, 4, 8 and 12 on a twelve-core host reduced the median build time from 19.32 s to 10.95 s, a speedup of only $1.76\times$ against an ideal of $12\times$, with parallel efficiency falling from 67% at two threads to 15% at twelve. The gap against the ideal line corresponds with Amdahl’s law [13]; the build ends in a `coalesce(1)` write that executes on a single thread regardless of the setting, and additional threads cannot reduce that cost.

5 Results & Discussion

All financial and margin figures are primarily computed for limited companies (AS) with financial statements in Norwegian kroner (NOK), restricted to companies with a computable operating margin. Three exceptions are made: Figures 5.6, 5.7, and 5.8 use all entity types. Figure 5.6 shows Stavanger’s top companies, including non-AS petroleum entities, Figure 5.7 ranks municipalities across the complete market population, and Figure 5.8 demonstrates how including non-AS entities such as Kongsberg Kraft IKS can distort profitability metrics, reinforcing the methodological decision to restrict most analyses to

limited companies.

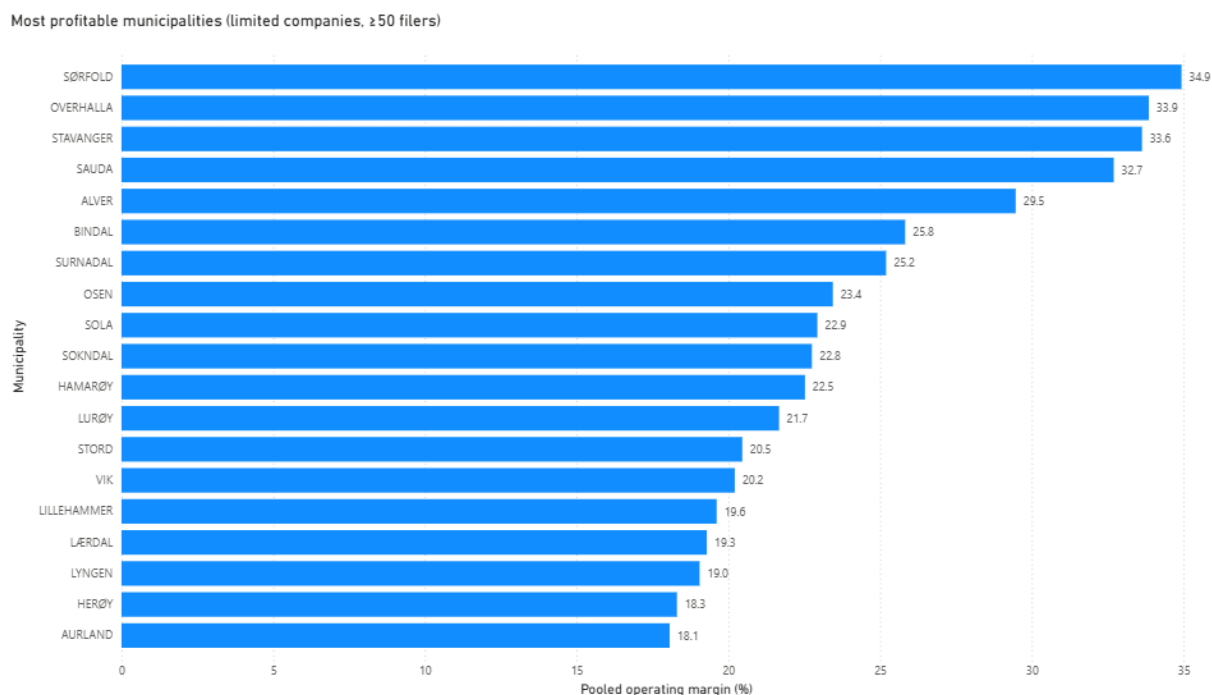


Figure 5.1: Most profitable municipalities among limited companies with at least fifty filing companies.

Evaluating municipal profitability requires distinguishing between total operating income (economic scale), pooled operating margin (revenue-weighted), and median operating margin (firm yield). To determine whether a municipality's financial performance comes from a single dominant company or broad economic health, we examine Figure 5.1. Ranking by pooled margin: Sørfold (34.9%), Overhalla (33.9%), Stavanger (33.6%), and Sauda (32.7%). This shows that the municipality's rank reflects high-revenue firms rather than the total company count.

Pooled vs. median operating margin by municipality

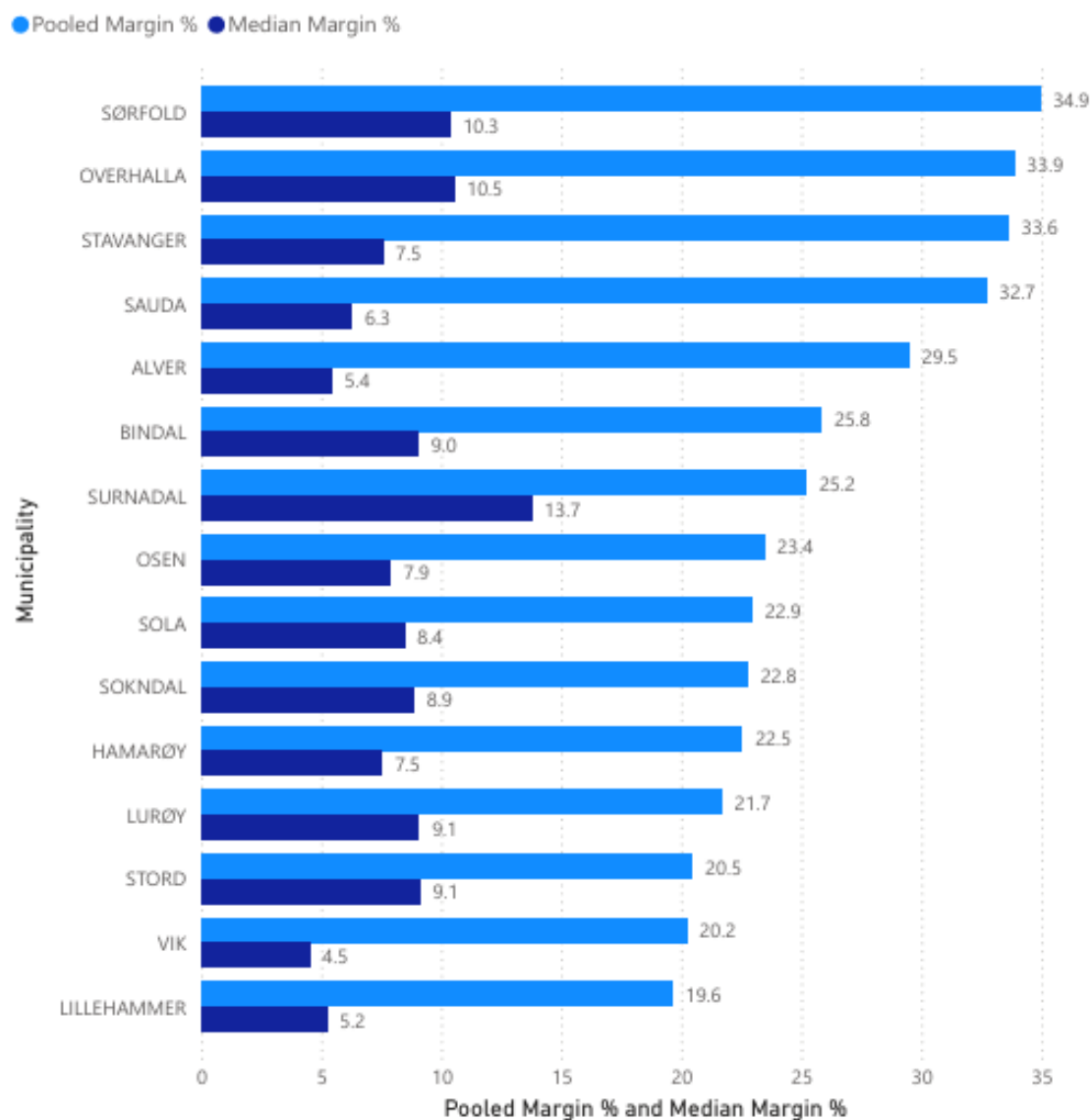


Figure 5.2: Pooled vs. median operating margin by municipality.

Figure 5.2 reveals that the pooled margin exceeds the median company's margin in every top municipality.

Municipality	Pooled Margin (%)	Median Margin (%)	Companies
SØRFOLD	34.93	10.34	61
OVERHALLA	33.85	10.50	154
STAVANGER	33.63	7.55	6341
SAUDA	32.70	6.27	194
Total	9.53	5.92	257202

Figure 5.3: Operating margins by municipality.

In Figure 5.3, we can see that high municipal profitability is driven by a small number of large companies rather than by broad local economies. Regardless of town size, the typical (median) company in these municipalities earns a 6-10% margin, yet the national median margin is 5.92%, and the national pooled margin is 9.53%, which reiterates the point that these few dominant companies bring in large revenue in these municipalities, giving the appearance that these municipalities are the most profitable.

Sørfold: companies by operating income (limited companies)

Company	Revenue (NOK)	Operating Income (NOK)	Operating Margin (%)
SISO ENERGI AS	283,865,984.00	181,644,652.00	63.99
SISOMAR AS	247,989,257.00	82,238,064.00	33.16
SALTEN HAVNESERVICE AS	43,672,367.00	4,255,228.00	9.74
ACUSTUS SVEIS AS	41,010,076.00	4,227,293.00	10.31
SALTEN ENTREPRENØR & EIENDOM AS	39,860,528.00	3,407,482.00	8.55
Total	862,392,844.00	301,238,368.00	34.93

Figure 5.4: Sørfold's companies by operating income (limited companies).

Sauda: companies by operating income (limited companies)

Company	Revenue (NOK)	Operating Income (NOK)	Operating Margin (%)
Aktieselskabet Saudefaldene	1,375,765,478.00	797,117,464.00	57.94
AARTUN EIENDOM AS	14,294,007.00	12,682,018.00	88.72
SAUDA AUTO AS	10,065,689.00	8,421,064.00	83.66
DJUVIK MASKINERING AS	45,881,825.00	7,217,846.00	15.73
EFFEKTIVT RENHOLD AS	68,134,207.00	6,527,629.00	9.58
Total	2,767,917,273.00	905,010,076.00	32.70

Figure 5.5: Sauda's companies by operating income (limited companies).

Stavanger: companies by operating income (all entity types)

Company	Revenue (NOK)	Operating Income (NOK)	Operating Margin (%)
TOTALENERGIES EP NORGE AS	61,500,000,000.00	40,740,000,000.00	66.24
A/S Norske Shell	25,249,000,000.00	20,514,000,000.00	81.25
OMV NORGE AS	16,747,422,000.00	8,722,268,000.00	52.08
LYSE PRODUKSJON AS	6,714,297,000.00	5,541,817,000.00	82.54
EQUINOR IN SALAH AS	5,057,474,374.00	3,680,249,969.00	72.77
Total	393,498,222,482.00	107,725,789,751.00	27.38

Figure 5.6: Stavanger's companies by operating income (all entity types).

The analyses confirm what drives each municipality: Stavanger, with large petroleum companies (Total Energies EP Norge AS, A/S Norske Shell; Equinor Energy AS is comparable in scale but reported in non-NOK currency and therefore excluded from the ranking); Sørfold with two companies (Siso Energi AS, Sisomar AS); Sauda with one company (Aktieselskabet Saudefaldene). This illustrates that the top-ranking municipalities come from anchor-firm concentration rather than broad economic health.

Most profitable municipalities-all entity

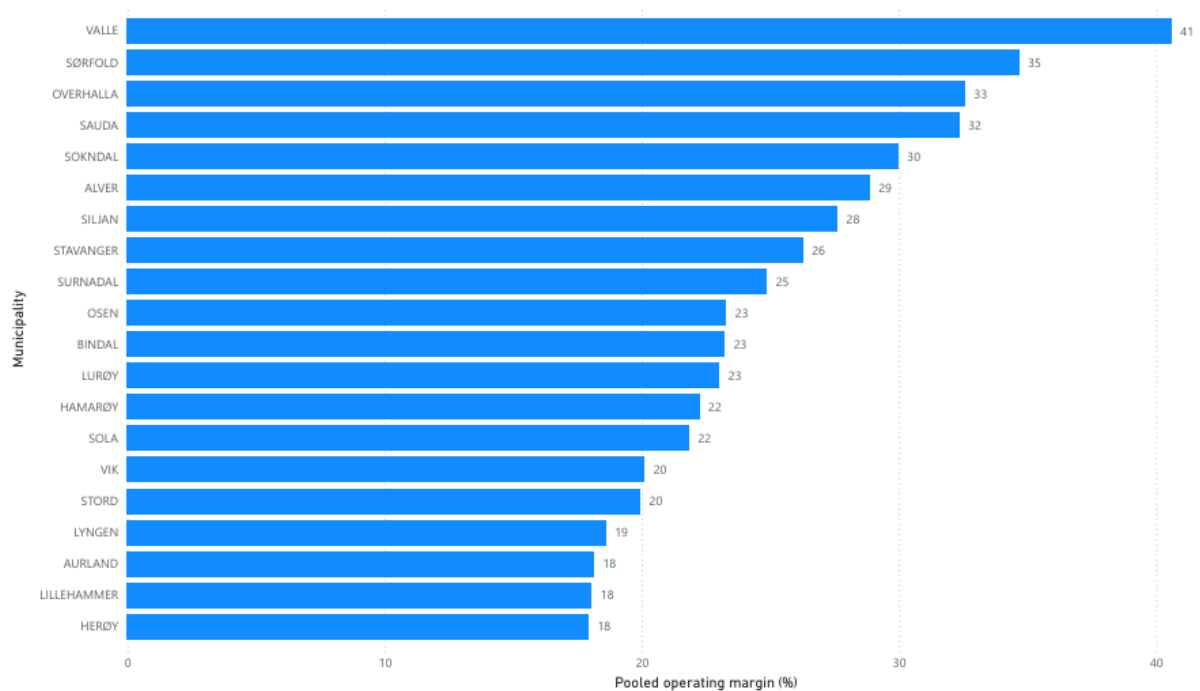


Figure 5.7: All-entity ranking.

Valle: companies by operating income (all entity types)

Company	Revenue (NOK)	Operating Income (NOK)	Operating Margin (%)
KONSESJONSKRAFT IKS	939,956,116.00	786,532,064.00	83.68
AREALEFFEKTIVE HYTTER AS	44,979,246.00	3,364,121.00	7.48
NOMELAND MASKIN AS	189,472,472.00	2,971,612.00	1.57
BRØDRENE TVEITEN MASKIN AS	20,278,520.00	2,925,589.00	14.43
NOMELAND PUKK AS	8,048,324.00	2,596,257.00	32.26
Total	1,741,287,682.00	708,439,291.00	40.68

Figure 5.8: Valle's companies by operating income (all entity types).

Further analyses show that revenue-weighting can produce distorted results when a dominant company is not an ordinary business. When expanding the scope to all entity types, Valle ranks first with a 41% pooled margin (Figure 5.7), yet the profitability comes mostly from one non-AS entity (Konsesjonskraft IKS). This single company accounts for 54% of total municipal revenue, with its operating income (786,532,064 NOK) greater than the entire municipality's aggregate (708,439,291 NOK). It reveals that the remainder of the local economy operates at a collective net loss. As its 84% margin reflects regulated resale of concession hydropower, not competitive market performance, filtering out non-AS entities is essential in municipal comparisons.

Operating margin by municipality (limited companies)

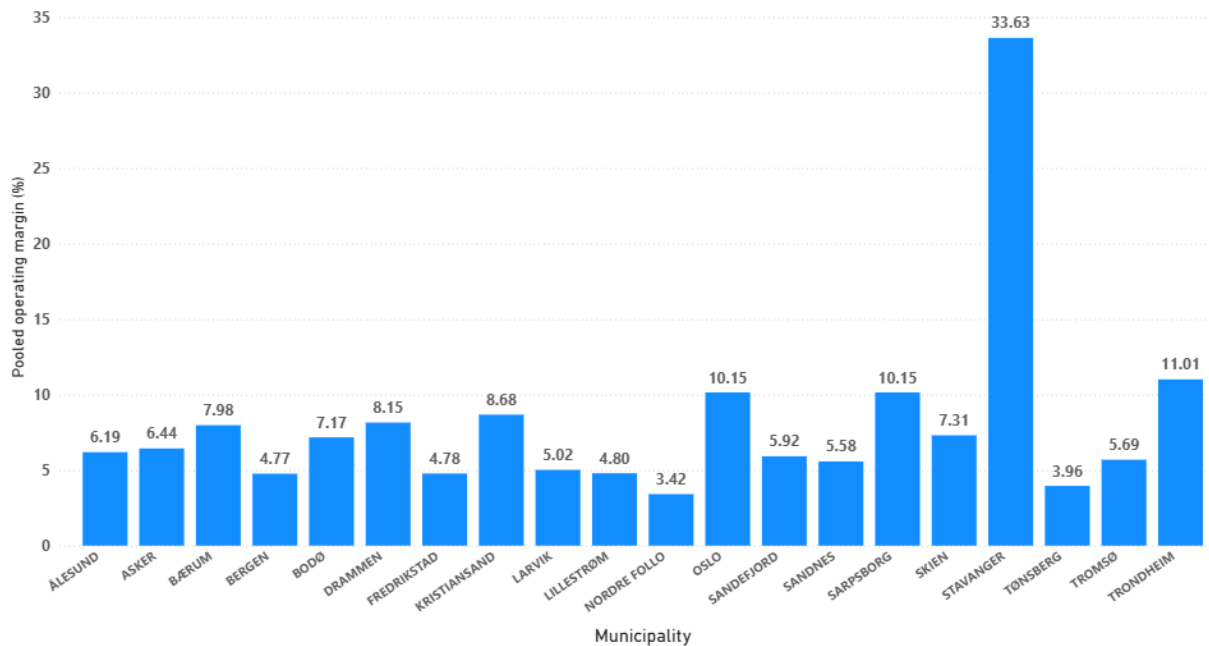


Figure 5.9: Pooled margin by municipality, limited companies only.

When restricting to limited companies, Valle disappears from the top ranking (Figure

5.9). This further validates the project’s decision to exclude non-AS entities, as it refines the analytical validity and prevents non-AS entities from distorting the metric.

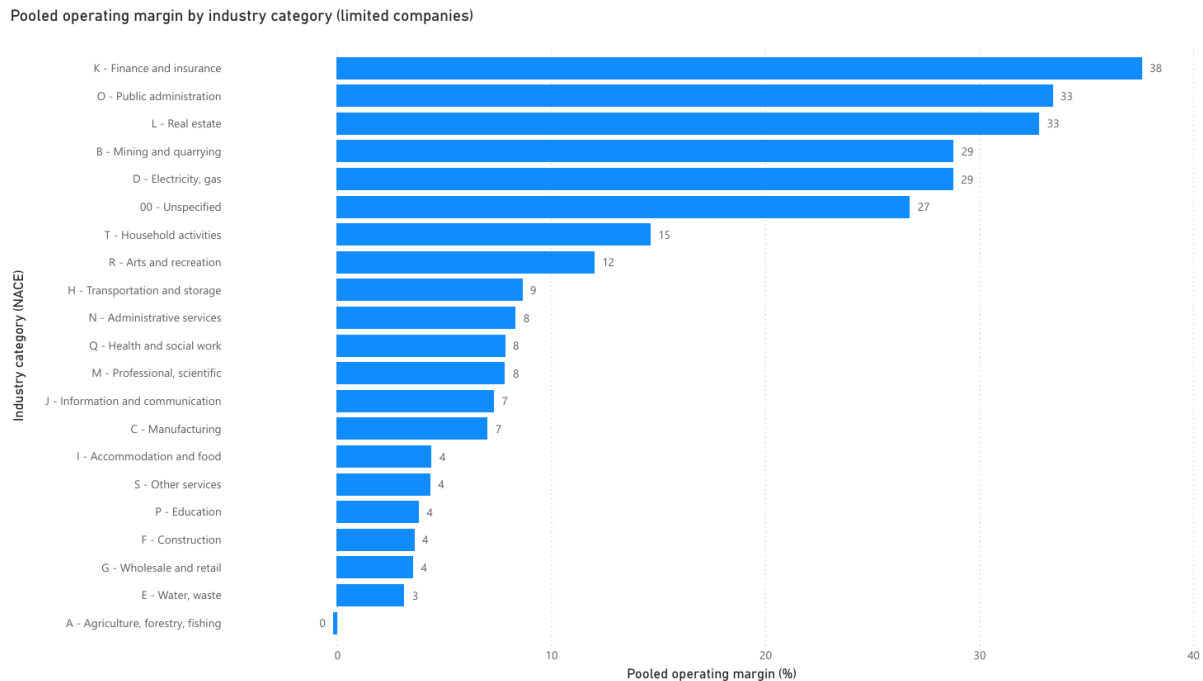


Figure 5.10: Pooled operating margin by NACE Rev. 2 (SN2007) section, limited companies.

To examine how profitability varies across macro industry categories, the pooled operating margin (total operating income $\div$ total revenue) was calculated for each industry category (NACE) Rev.2 (SN2007).

The three highest-ranking sectors are K - Finance and insurance (38%), O – Public administration (33%), and L – Real estate (33%). However, they are inflated by non-operational income, B – Mining and quarrying and D – Electricity and gas (both 29%). in operational profitability. The "00 – Unspecified" category (27%), which consists of holding companies not assigned a NACE code, ranks similarly high due to investment returns with minimal operating costs. At the far end, Agriculture records a pooled margin of zero.

Public administration: companies by operating income

Company	Revenue (NOK)	Operating Income (NOK)	Operating Margin (%)
FERDE AS	5,097,212,008.00	4,797,181,271.00	94.1
FJELLINJEN AS	5,014,371,000.00	4,666,246,000.00	93.1
VEGAMOT AS	2,493,794,675.00	2,363,175,901.00	94.8
INNLANDET ENERGI HOLDING AS	695,331,242.00	691,833,245.00	99.5
BPS NORD - BYPAKKE TENK TROMSØ AS	277,580,348.00	241,897,112.00	87.1
Total	42,984,254,050.00	14,380,432,269.00	33.5

Figure 5.11: Companies in the Public administration category.

Toll road companies with high margins (like Ferde AS (94.11%), Fjellinjen AS (93.06%), and Vegamot AS (94.76%)) dominate the “Public administration” category. These companies collect revenue from roads that are already built; hence, the daily operating costs are low. This results in significantly high margins that come from regulated monopolies rather than competitive commercial performance.

Real estate: companies by operating income

Company	Revenue (NOK)	Operating Income (NOK)	Operating Margin (%)
HELGELAND INVEST AS	1,732,882,320.00	1,725,174,322.00	99.6
FERNCLIFF TIH AS	1,439,239,833.00	1,428,370,177.00	99.2
FERNCLIFF HOLDING AS	1,390,833,000.00	1,388,162,000.00	99.8
GJELSTEN HOLDING AS	1,400,968,000.00	1,248,152,000.00	89.1
NORDEA LIV EIENDOM AS	605,434,688.00	1,245,191,571.00	205.7
Total	357,434,169,791.00	117,326,572,297.00	32.8

Figure 5.12: Companies in the Real Estate category.

The Real Estate category is dominated by property-holding firms, with several large entities reporting high operating margins, leading by Nordea Liv Eiendom AS (205.67%) exceeding over 100%, and following by Helgeland Invest AS (99.56%), Ferncliff Holding AS (99.81%), Ferncliff Tih AS (99.24%), and Gjelsten Holding AS (89.09%) being near 100%, as shown in Figure 5.12. The total pooled margin is 32.82%, and when a company’s margin exceeds 100% (Nordea), it indicates that the company earns a significant amount of income outside its rental sales, making its operating income larger than its revenue. Under Norwegian Generally Accepted Accounting Principles (NGAAP), companies must follow the realisation principle [8, Section 4-1, nr. 2], meaning that income is only recognised when earned. Because assets are measured at acquisition cost [8, Section 5-3], unsold

property value increases are excluded from profit or loss. Therefore, the high margins are not due to high tenant rents. Instead, these high margins can be from other factors such as selling property for a realised gain, charging internal management fees, or applying fair-value adjustments under IFRS rules [14].

Finance: companies by operating income

Company	Revenue (NOK)	Operating income (NOK)	Operating Margin %
FERD HOLDING AS	5,350,000,000.00	5,342,000,000.00	99.9
DNB ASSET MANAGEMENT AS	3,180,545,000.00	1,635,268,000.00	51.4
HV CAPITAL VII AS	1,403,256,962.00	1,367,894,552.00	97.5
SALT VALUE AS	1,373,118,140.00	1,052,811,481.00	76.7
ARGENTUM FONDSINVESTERINGER AS	1,117,993,000.00	1,018,220,000.00	91.1
CTS CAPITAL AS	950,856,794.00	942,070,676.00	99.1
OM HOLDING AS	870,438,805.00	870,336,752.00	100.0
DNB ASSET MANAGEMENT HOLDING AS	870,000,000.00	869,852,000.00	100.0
ARGENTUM NPEP AS	728,583,213.00	727,603,355.00	99.9
STOREBRAND ASSET MANAGEMENT AS	3,279,000,000.00	699,000,000.00	21.3
EFØY INVESTERING AS	631,931,160.00	620,407,214.00	98.2
PARETO SECURITIES AS	1,851,400,000.00	599,200,000.00	32.4
MATTISBERGET AS	497,750,153.00	491,260,549.00	98.7
HITECVISION AS	498,446,697.00	488,196,473.00	97.9
OSLO PENSJONSFORSIKRING AS	18,802,000,000.00	470,000,000.00	2.5
Total	92,595,018,736.00	34,828,775,022.00	37.6

Figure 5.13: Companies in the Finance category.

In Figure 5.13, we see that high margins are driven by investment holding companies such as OM Holding AS (100%), DNB Asset Management Holding AS (100%), and Ferd Holding AS (99.9%). These companies have near-perfect margins because their operating income comes from investment returns while the operating expenses are low. Conversely, Oslo Pensjonsforsikring AS (2.5%) is a life insurer. The low margin is because most of the revenue must be set aside in reserve to cover future pension payouts. Security brokers like Pareto Securities AS (32.4%) are a real financial business, as the company is an active brokerage firm where client fees offset operating expenses. In conclusion, the finance sector's high margin is mainly owing to holding companies, which inflate the metric.

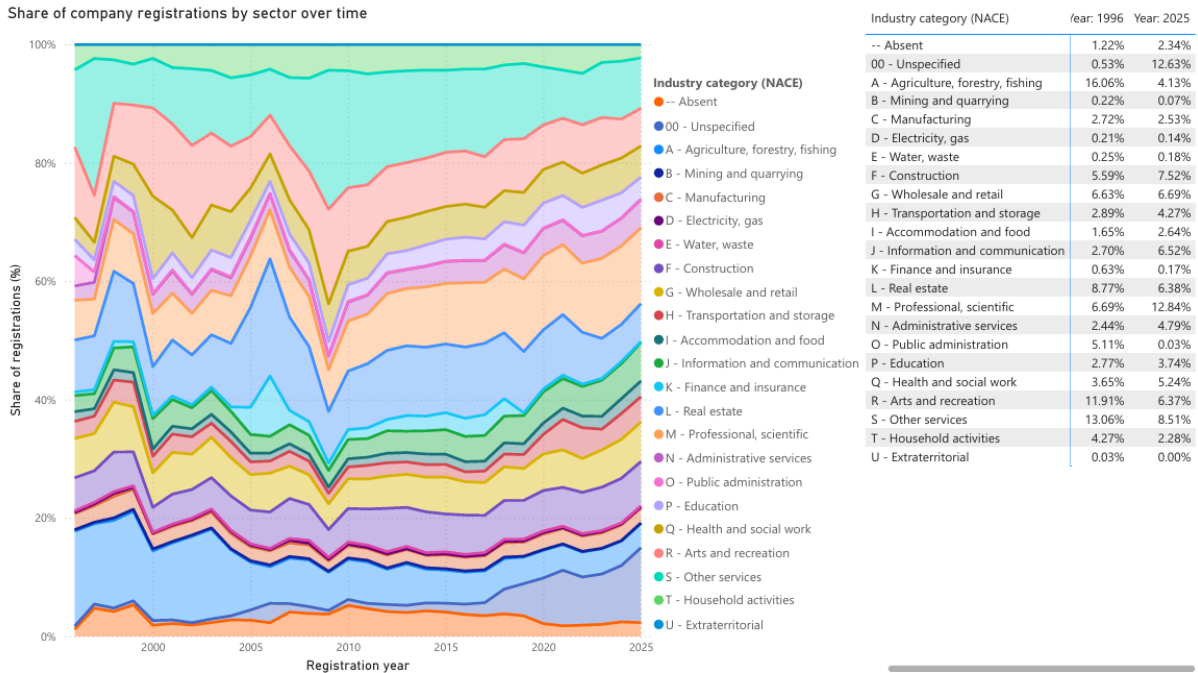


Figure 5.14: Share of company registrations by NACE section (SN2007), 1996–2025. Left: sector composition over time; right: exact shares at the start and end of the period.

The years 1995 and 2026 are excluded from this analysis as 1995 aggregates all historical registrations prior to that date, creating an anomaly in the data, while 2026 is partially recorded. Since the dataset only includes active entities, evaluating counts introduced survivorship bias. To reduce this, we converted company registrations into annual percentages per sector from 1996 to 2025.

Figure 5.14 shows the largest sectoral gain is M – Professional and scientific services, which rose from 6.69% to 12.84%, followed by J – Information and communication (2.70% to 6.52%). Other growing sectors include Q – Health and social work (3.65% to 5.24%), F – Construction (5.59% to 7.52%), and N – Administrative services (2.5% to 4.8%). The steepest declines are A – Agriculture, forestry and fishing, which fell from 16.06% to 4.13%, and S – Other services (13.06% to 8.51%), alongside L – Real estate (8.77% to 6.38%) and R – Arts and recreation (11.91% to 6.37%). A sharp growth of the category “00 – Unspecified”, from 0.53% to 12.63% of holding companies are new companies. Around 2018-2019, the share of section K - Finance and insurance dropped while the “00 - Unspecified” rises corresponding, suggesting a change in how newly registered companies are handled rather than a real economic shift.

To evaluate market saturation and competitor density, the number of registered companies per 1000 residents was calculated for each municipality in two sectors: accommo-

dation (NACE 55) and food (NACE 56) and grocery stores (NACE 47.110). The analysis was restricted to municipalities of at least 5,000 residents to avoid per-capita volatility.

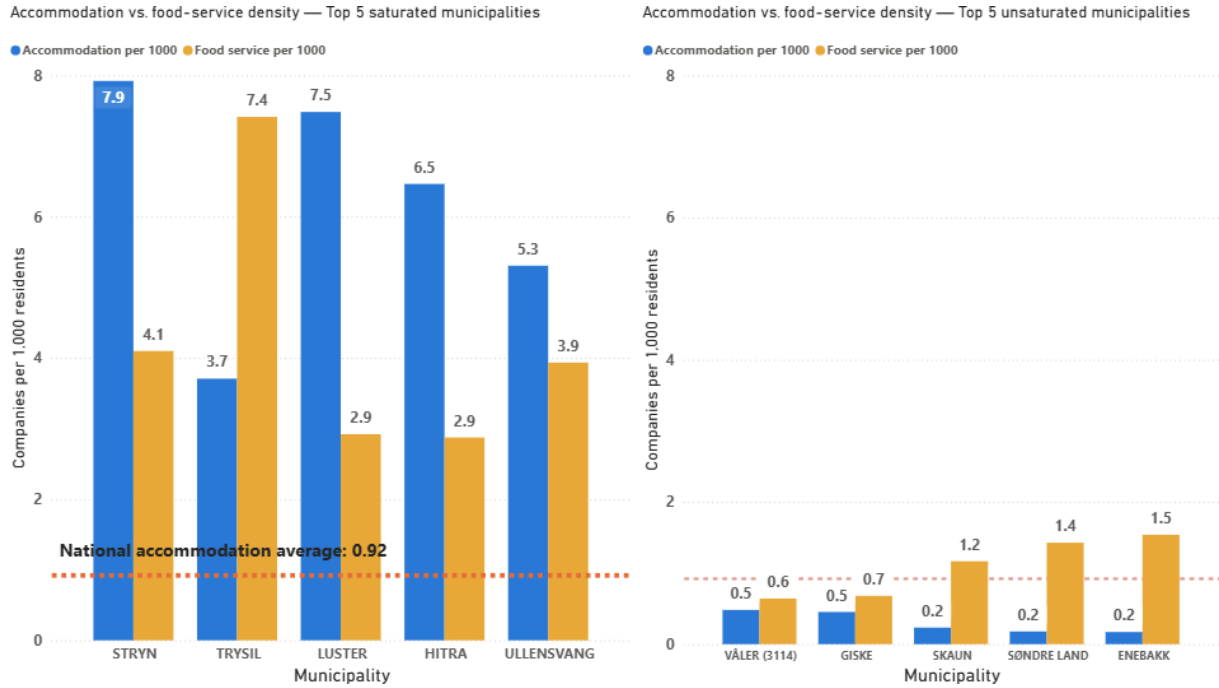


Figure 5.15: Accommodation vs. food-service density per 1,000 residents. Left: the top 5 most saturated; Right: top 5 least unsaturated municipalities.

In Figure 5.15, accommodation and food-service density varies geographically, ranging from 1.1 to 12.0 companies per 1,000 residents. The five most saturated municipalities are Stryn (7.9 accommodation, 4.1 food service), Trysil (3.7, 7.4), Luster (7.5, 2.9), Hitra (6.5, 2.9), and Ullensvang (5.3, 3.9). In contrast, the least-served municipalities are Våler (0.5, 0.6), Giske (0.5, 0.7), Skaun (0.2, 1.2), Søndre Land (0.2, 1.4), and Enebakk (0.2, 1.5). Comparing these municipalities to the national average of 0.92 accommodation companies per 1,000 residents, the top five all sit above this threshold, while the bottom five all fall below it. The consistency of high numbers in the two categories points to tourism. However, this remains an inference as the dataset tracks registered business entities rather than consumers.

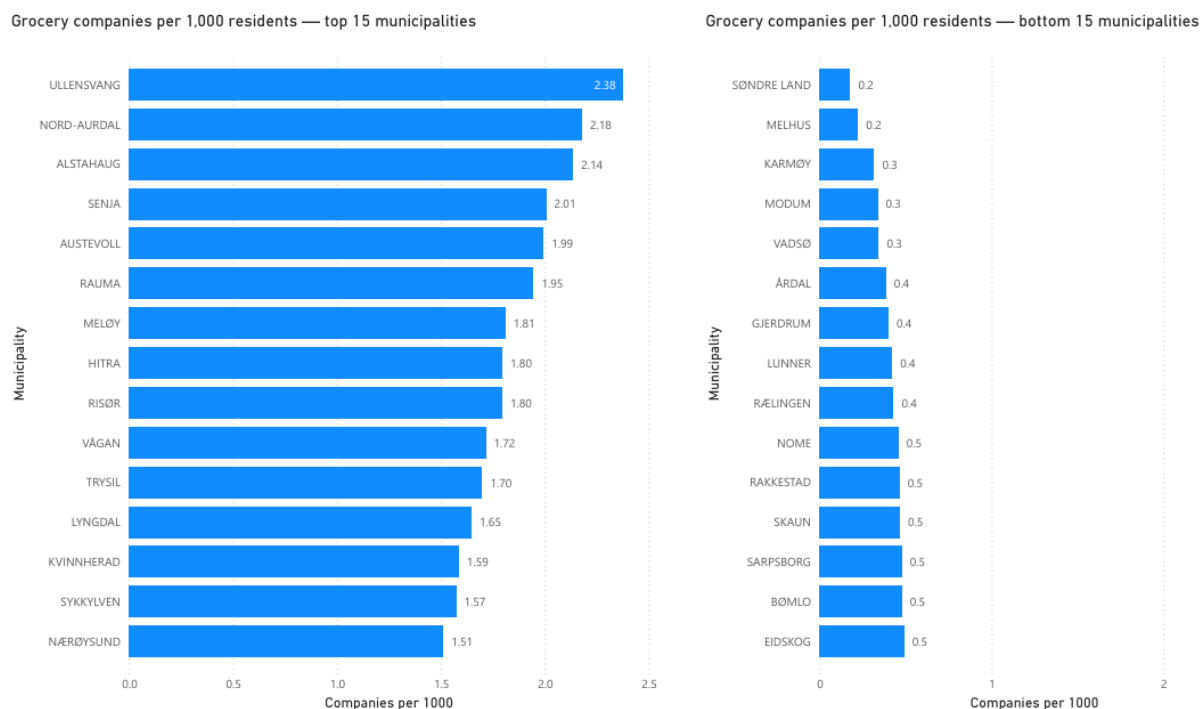


Figure 5.16: Grocery stores per 1,000 residents; by municipality. Left: the 15 most grocery-dense municipalities; Right: the 15 lowest density.

Grocery density spans a narrow range of 0.2 to 2.4 companies per 1,000 residents, compared to 1.1 to 12.0 for food service and accommodation. The most grocery-dense municipality (Ullensvang, 2.38) reaches barely a fifth of the top food-service peak (Stryn, 12.0). Combined with Figure 5.18, this scale shows that food service is consistent with reliance on tourism, whereas grocery stores scale directly with residents.

6 Conclusions

This project transformed over 1.1 million records from Norway official company register[4] and enriched the company data with financial records [7] and population data on the municipality level [15] by using MongoDB, Apache Spark, and Microsoft Power BI. The resulting pipeline converts raw data into a set of Power BI dashboards that allow investors, managers, and other stakeholders to evaluate municipal and sector dynamics.

We discovered that a municipality or sector can look profitable when the figure is driven by a single company or non-operating entity. A municipality can appear richer due to the pooled averages, and therefore, it is best to look at the median operating margins, which

reduces the effect of extreme outliers and skewed data. High and genuine profits come from major industries such as hydropower, fish farming, and oil and gas. In addition, struggling companies leave financial indicators before they collapse. Beyond profitability, the density analysis showed that market saturation is sector-dependent.

In the future, it would be useful to incorporate a multi-year financial data source to test early bankruptcy indicators, including deregistered companies, which allows the registration trends to be measured without survivorship bias, and external data to confirm inferences from the density analysis.

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